Greeks by Monte Carlo: Bumping, Pathwise, and Likelihood Ratio

Quant Finance Portfolio — Phase 2, Block 4

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Abstract

We compute the European-call Greeks (Delta, Vega, Gamma) by Monte Carlo using three increasingly sophisticated techniques: bumping (finite differences applied to the pricer), pathwise sensitivities (differentiation of the payoff under the integral), and likelihood ratio (differentiation of the kernel of the underlying density). All three are unbiased for the appropriate Greeks under appropriate regularity conditions; they trade off variance, bias, and applicability differently. We derive each from first principles, implement them under the exact GBM sampler of Block 1.1, and validate against closed-form Black-Scholes Greeks at the canonical ATM point. The block also surfaces a delicate but important point: the pathwise method does *not* apply to Gamma (the kink of $\max(\cdot, 0)$ produces a Dirac contribution after the second derivative), and the likelihood-ratio method is known to be high-variance for Gamma; in practice Gamma is computed by central finite differences.

Contents

1 The setting **1**

2 Bumping (finite differences) **1**
- 2.1 Construction . . . 2
- 2.2 Common random numbers (CRN) . . . 2
- 2.3 The bias-variance trade-off in h . . . 2

3 Pathwise sensitivities **2**
- 3.1 The interchange of differentiation and expectation . . . 3
- 3.2 Pathwise Delta and Vega for the European call . . . 3
- 3.3 Why pathwise fails for Gamma . . . 3

4 Likelihood ratio **4**
- 4.1 The construction . . . 4
- 4.2 Score function for GBM . . . 4
- 4.3 Variance comparison: pathwise vs LR . . . 5

5 Algorithmic specifications **5**

6 Empirical validation strategy **6**

7 Bibliographic notes **6**

1 The setting

The price of a European call under risk-neutral GBM is

$$C(S_0,\sigma) \;=\; e^{-rT}\,\mathbb{E}\big[\max(S_T-K,0)\big], \qquad S_T \;=\; S_0\exp\big((r-\tfrac{1}{2}\sigma^2)T+\sigma\sqrt{T}\,Z\big), \qquad Z\sim\mathcal{N}(0,1). \tag{1}$$

The Greeks of interest are

$$\Delta \;=\; \frac{\partial C}{\partial S_0}, \qquad \nu \;=\; \frac{\partial C}{\partial \sigma}, \qquad \Gamma \;=\; \frac{\partial^2 C}{\partial S_0^2}. \tag{2}$$

Black-Scholes provides closed forms (Block 1):

$$\Delta^{\mathrm{BS}} = \Phi(d_1), \qquad \nu^{\mathrm{BS}} = S_0\phi(d_1)\sqrt{T}, \qquad \Gamma^{\mathrm{BS}} = \frac{\phi(d_1)}{S_0\sigma\sqrt{T}}, \tag{3}$$

with ϕ the standard normal pdf and $d_1 = (\log(S_0/K) + (r+\sigma^2/2)T)/(\sigma\sqrt{T})$. We use these as ground truth.

The MC pricer of Block 1.1 produces an estimate $\hat{C}_n = \frac{1}{n}\sum_{i=1}^n e^{-rT}\max(S_T^{(i)}-K,0)$ where $S_T^{(i)} = S_0\exp((r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z_i)$ for Z_i iid $\mathcal{N}(0,1)$. Greeks are MC estimators of $\partial C/\partial\theta$ for $\theta\in\{S_0,\sigma\}$ or of the second derivative.

2 Bumping (finite differences)

The conceptually simplest Greek estimator: differentiate the *pricer*, not the payoff.

2.1 Construction

Let $\hat{C}_n(\theta)$ be the MC estimate at parameter θ. Choose a small bump h. The forward-difference estimator of $\partial C/\partial\theta$ is

$$\widehat{\partial_\theta C}_n^{\mathrm{FD}} \;=\; \frac{\hat{C}_n(\theta+h)-\hat{C}_n(\theta)}{h}. \tag{4}$$

The central-difference estimator is

$$\widehat{\partial_\theta C}_n^{\mathrm{CD}} \;=\; \frac{\hat{C}_n(\theta+h)-\hat{C}_n(\theta-h)}{2h}. \tag{5}$$

For Gamma we apply central differences twice:

$$\widehat{\Gamma}_n \;=\; \frac{\hat{C}_n(S_0+h)-2\hat{C}_n(S_0)+\hat{C}_n(S_0-h)}{h^2}. \tag{6}$$

2.2 Common random numbers (CRN)

Naively, the two evaluations $\hat{C}_n(\theta+h)$ and $\hat{C}_n(\theta-h)$ would use independent Monte Carlo samples. The variance of the difference is then $\mathrm{Var}(\hat{C}_n(\theta+h))+\mathrm{Var}(\hat{C}_n(\theta-h)) \approx 2\sigma^2/n$, and dividing by $2h$ amplifies this further. The result: a bump-and-revalue Greek has variance scaling as $\sigma^2/(nh^2)$, which blows up as $h\to 0$.

The standard fix is *common random numbers*: use the same set of $\{Z_i\}$ for both evaluations. Pathwise on each i, the only thing that changes between the two evaluations is the parameter, not the noise. The variance of the difference is then much smaller because the noise mostly cancels. For our setting,

$$\mathrm{Var}\Big(\hat{C}_n(S_0+h,\{Z_i\})-\hat{C}_n(S_0-h,\{Z_i\})\Big) \;\approx\; h^2\cdot\mathrm{Var}(\partial_{S_0}f_i) \tag{7}$$

to leading order in h, where $f_i = e^{-rT}\max(S_T^{(i)}-K,0)$. The factor of h^2 cancels with the $1/h^2$ in Γ, leaving a finite variance estimator.

2.3 The bias-variance trade-off in h

With CRN, the bumping estimator has:

- Bias: $O(h)$ for forward differences, $O(h^2)$ for central differences.
- Variance: roughly constant in h (after CRN cancellation).

The optimal h minimises bias^2 + variance. For central differences with bias $O(h^2)$ and variance $O(1)$, the optimum is where both contribute equally, giving $h_\star$ on the order of $\epsilon^{1/4}$ where ϵ is the floating-point precision ($\sim 10^{-16}$), so $h_\star \sim 10^{-4}$ for double precision. We use $h = 10^{-2}$ for stability against MC noise; the bias is then $\sim 10^{-4}$, comfortably below the half-width at our budgets.

For Gamma, the second-order central difference has bias $O(h^2)$ but variance $O(1/h^2)$ even with CRN (the leading-order cancellation that worked for Δ does not extend to the second derivative). Gamma is intrinsically harder.

3 Pathwise sensitivities

The bumping method differentiates the pricer. The pathwise method differentiates the *payoff*.

3.1 The interchange of differentiation and expectation

Suppose $f(\theta, Z)$ is the payoff (a function of the parameter θ and the underlying randomness Z), and we want to estimate $\partial_\theta \mathbb{E}[f(\theta, Z)]$. If f is sufficiently regular, we can interchange differentiation and expectation:

$$\frac{\partial}{\partial\theta}\mathbb{E}[f(\theta, Z)] \;=\; \mathbb{E}\left[\frac{\partial f(\theta, Z)}{\partial\theta}\right]. \tag{8}$$

The Monte Carlo estimator of the right-hand side is

$$\widehat{\partial_\theta C_n}^{\mathrm{PW}} \;=\; \frac{1}{n}\sum_{i=1}^{n}\frac{\partial f(\theta, Z_i)}{\partial\theta}. \tag{9}$$

This is unbiased for the true derivative if the interchange is justified, which by dominated convergence requires f Lipschitz in θ uniformly in Z (or weaker conditions). The European call payoff $\max(S_T - K, 0)$ is Lipschitz in S_T (with constant 1), which is in turn C^∞ in S_0 and σ, so the interchange is valid.

3.2 Pathwise Delta and Vega for the European call

Compute $\partial f/\partial S_0$ first. Using

$$S_T = S_0 \exp\big((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z\big) \;=\; \frac{S_T}{S_0}\cdot S_0, \tag{10}$$

so $\partial S_T/\partial S_0 = S_T/S_0$. By the chain rule (and noting that $\max(x, 0)$ has derivative $\mathbf{1}_{\{x>0\}}$ almost everywhere),

$$\frac{\partial f}{\partial S_0} \;=\; e^{-rT}\cdot \mathbf{1}_{\{S_T>K\}}\cdot\frac{S_T}{S_0}. \tag{11}$$

For Vega:

$$\frac{\partial S_T}{\partial\sigma} \;=\; S_T\cdot(-\sigma T + \sqrt{T}\,Z), \tag{12}$$

so

$$\frac{\partial f}{\partial \sigma} \;=\; e^{-rT} \cdot \mathbf{1}_{\{S_T > K\}} \cdot S_T \cdot (\sqrt{T}\, Z - \sigma T). \tag{13}$$

The pathwise Delta and Vega estimators are then

$$\widehat{\Delta}_n^{\mathrm{PW}} \;=\; \frac{1}{n} \sum_{i=1}^{n} e^{-rT} \mathbf{1}_{\{S_T^{(i)} > K\}} \frac{S_T^{(i)}}{S_0}, \tag{14}$$

$$\widehat{\nu}_n^{\mathrm{PW}} \;=\; \frac{1}{n} \sum_{i=1}^{n} e^{-rT} \mathbf{1}_{\{S_T^{(i)} > K\}} S_T^{(i)} (\sqrt{T}\, Z_i - \sigma T). \tag{15}$$

These are unbiased and have variance comparable to (often less than) the variance of the price estimator itself. They are the gold standard when applicable.

3.3 Why pathwise fails for Gamma

For Gamma we want $\partial^2 C / \partial S_0^2$, which would require differentiating $\partial f / \partial S_0$ once more. Computing:

$$\frac{\partial^2 f}{\partial S_0^2} \;=\; \frac{\partial}{\partial S_0} \big[e^{-rT} \mathbf{1}_{\{S_T > K\}} \frac{S_T}{S_0} \big]. \tag{16}$$

The indicator $\mathbf{1}_{\{S_T > K\}}$ is the Heaviside step function of $S_T - K$. Its derivative in the distributional sense is the Dirac delta $\delta(S_T - K)$. Once we differentiate, the expectation

$$\mathbb{E}[\delta(S_T - K) \cdot (\text{stuff})] \tag{17}$$

is finite (it equals (stuff at $S_T = K$) times the density of S_T at K), but *not estimable by Monte Carlo*: a single sample has $S_T = K$ with probability zero, so the sum is identically zero on every realisation.

This is the canonical failure of pathwise for second-order Greeks of payoffs with kinks: the second derivative is a measure, not a function, and Monte Carlo can only estimate functions.

4 Likelihood ratio

The likelihood-ratio (LR) method, developed by Broadie and Glasserman, sidesteps the problem of differentiating the payoff by differentiating the *density of the underlying* instead.

4.1 The construction

Suppose S_T has a density $p(s;\theta)$ depending on the parameter θ. Then

$$C(\theta) \;=\; e^{-rT} \int f(s)\, p(s;\theta)\, \mathrm{d}s, \tag{18}$$

and (assuming we can interchange differentiation and integration)

$$\frac{\partial C}{\partial \theta} \;=\; e^{-rT} \int f(s) \cdot \frac{\partial p(s;\theta)/\partial \theta}{p(s;\theta)} \cdot p(s;\theta)\, \mathrm{d}s \;=\; e^{-rT} \mathbb{E}[f(S_T) \cdot \mathcal{S}(\theta, S_T)]\,, \tag{19}$$

where

$$\mathcal{S}(\theta, S_T) \;\stackrel{\text{def}}{=}\; \frac{\partial}{\partial \theta} \log p(S_T;\theta) \tag{20}$$

is the *score function* associated with the density family.

The MC estimator is

$$\widehat{\partial_\theta C}_n^{\mathrm{LR}} \;=\; \frac{1}{n} \sum_{i=1}^{n} e^{-rT} f(S_T^{(i)}) \cdot \mathcal{S}(\theta, S_T^{(i)}). \tag{21}$$

4.2 Score function for GBM

Under GBM, $\log S_T = \log S_0 + (r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z$ is normal with mean $\mu = \log S_0 + (r - \frac{1}{2}\sigma^2)T$ and standard deviation $\nu = \sigma\sqrt{T}$. The density of $\log S_T$ is

$$p_{\log}(\ell; S_0, \sigma) \;=\; \frac{1}{\nu\sqrt{2\pi}} \exp\left(-\frac{(\ell-\mu)^2}{2\nu^2}\right). \tag{22}$$

Differentiating $\log p_{\log}$ with respect to S_0:

$$\mathcal{S}_{S_0}(\log S_T) \;=\; \frac{Z}{S_0\sigma\sqrt{T}}. \tag{23}$$

With respect to σ:

$$\mathcal{S}_{\sigma}(\log S_T) \;=\; \frac{Z^2-1}{\sigma} - Z\sqrt{T}. \tag{24}$$

(Both derivations use the chain rule and the fact that $Z = (\log S_T - \mu)/\nu$.)

The LR Delta and Vega estimators are

$$\widehat{\Delta}_n^{\mathrm{LR}} \;=\; \frac{1}{n}\sum_{i=1}^{n} e^{-rT}\max(S_T^{(i)} - K, 0)\cdot\frac{Z_i}{S_0\sigma\sqrt{T}}, \tag{25}$$

$$\widehat{\nu}_n^{\mathrm{LR}} \;=\; \frac{1}{n}\sum_{i=1}^{n} e^{-rT}\max(S_T^{(i)} - K, 0)\cdot\left(\frac{Z_i^2-1}{\sigma} - Z_i\sqrt{T}\right). \tag{26}$$

4.3 Variance comparison: pathwise vs LR

For the European call, both pathwise and LR are unbiased. The variance comparison is:

- Pathwise: the estimand is $e^{-rT}\mathbf{1}_{\{S_T>K\}}(S_T/S_0)$, which is bounded for typical paths and has moderate variance.
- LR: the estimand is $f(S_T)\cdot Z/(S_0\sigma\sqrt{T})$, which carries the full payoff f multiplied by a factor that can have large absolute value when $|Z|$ is large. Tails of Z contribute $f(S_T)\cdot Z$, which can be large.

Empirically, for the call, pathwise has variance roughly 2–3× smaller than LR. The advantage of LR is universality: for digital options where f is a Heaviside (not Lipschitz), pathwise fails but LR works.

5 Algorithmic specifications

Algorithm 1 Bump-and-revalue Delta with central differences and CRN

Require: Spot S_0, parameters, sample size n, bump h, rng.

1: Generate $\{Z_i\}_{i=1}^n$ iid $\mathcal{N}(0,1)$.
2: $C^+ \leftarrow \frac{1}{n}\sum_i e^{-rT}\max(S_0(1+h)e^{(r-\sigma^2/2)T+\sigma\sqrt{T}Z_i} - K, 0)$
3: $C^- \leftarrow \frac{1}{n}\sum_i e^{-rT}\max(S_0(1-h)e^{(r-\sigma^2/2)T+\sigma\sqrt{T}Z_i} - K, 0)$
4: $\Delta \leftarrow (C^+ - C^-)/(2S_0h)$
5: **return** Δ

The trick: we bump S_0 multiplicatively (by $1 \pm h$) so that the formulae work uniformly without recomputing the diffusion coefficient.

Algorithm 2 Pathwise Delta

Require: Spot S_0, parameters, sample size n, rng.

Generate $\{Z_i\}_{i=1}^n$ iid $\mathcal{N}(0,1)$.
for $i = 1, \ldots, n$ **do**
  $S_T^{(i)} \leftarrow S_0 \exp((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i)$
  $\Delta_i \leftarrow e^{-rT} \cdot \mathbf{1}_{\{S_T^{(i)} > K\}} \cdot S_T^{(i)}/S_0$
end for
return $\bar{\Delta} = n^{-1} \sum_i \Delta_i$

Algorithm 3 Likelihood-ratio Delta

Require: Spot S_0, parameters, sample size n, rng.

Generate $\{Z_i\}_{i=1}^n$ iid $\mathcal{N}(0,1)$.
for $i = 1, \ldots, n$ **do**
  $S_T^{(i)} \leftarrow S_0 \exp((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i)$
  $\Delta_i \leftarrow e^{-rT} \max(S_T^{(i)} - K, 0) \cdot Z_i/(S_0\sigma\sqrt{T})$
end for
return $\bar{\Delta} = n^{-1} \sum_i \Delta_i$

6 Empirical validation strategy

The script `python/validate_greeks_mc.py` runs four tests.

Test 1: BS coherence. For each Greek and each method, the estimator at $n = 10^5$ should agree with the closed-form BS Greek within a few standard errors.

Test 2: variance comparison Delta. Pathwise vs LR vs bumping. Predicted: pathwise $<$ LR $<$ bumping. Report variance ratios at fixed budget.

Test 3: variance comparison Vega. Same as Test 2.

Test 4: input validation. Mechanical: bad inputs should raise.

The Block 4 implementation is a single new module `quantlib.greeks` containing nine pricers (3 methods $\times$ 3 Greeks, with the caveat that pathwise/LR Gamma is "not implemented" and falls back to bumping).

7 Bibliographic notes

The pathwise method for option Greeks is due to Broadie and Glasserman [1]. The likelihood-ratio method is also from [1]; the broader context of efficient simulation of derivatives is in [3, Chapter 7]. The malliavin calculus generalisation, which provides a unified framework for both pathwise and LR, is in Fournié et al. [2]. We do not pursue the Malliavin generalisation in this writeup but note that it is the modern theoretical lens.

References

[1] M. Broadie and P. Glasserman, *Estimating security price derivatives using simulation*, Management Science, 42(2):269–285, 1996.

[2] E. Fournié, J. M. Lasry, J. Lebuchoux, P. L. Lions, N. Touzi, *Applications of Malliavin calculus to Monte-Carlo methods in finance*, Finance and Stochastics, 3(4):391–412, 1999.

[3] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.